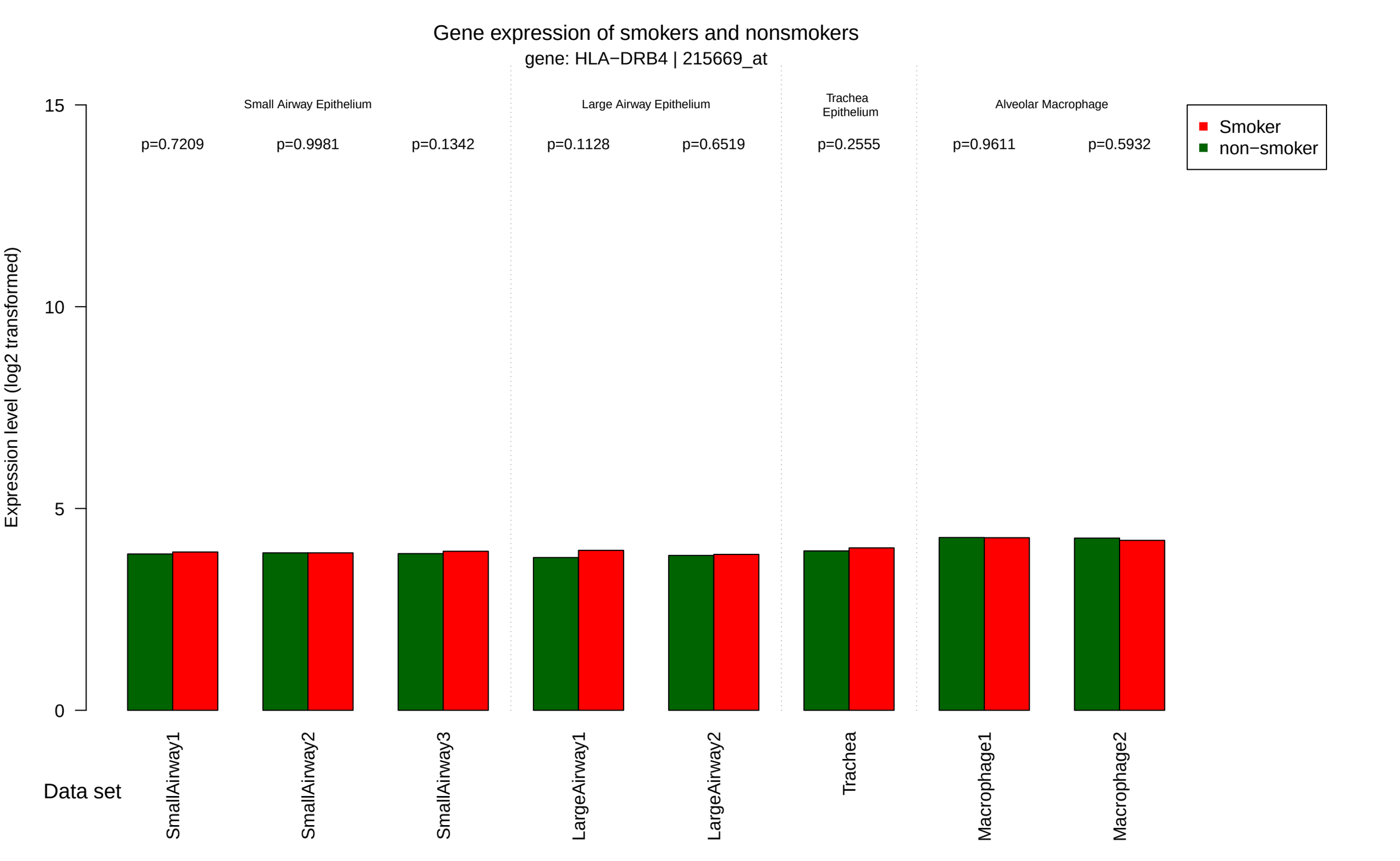
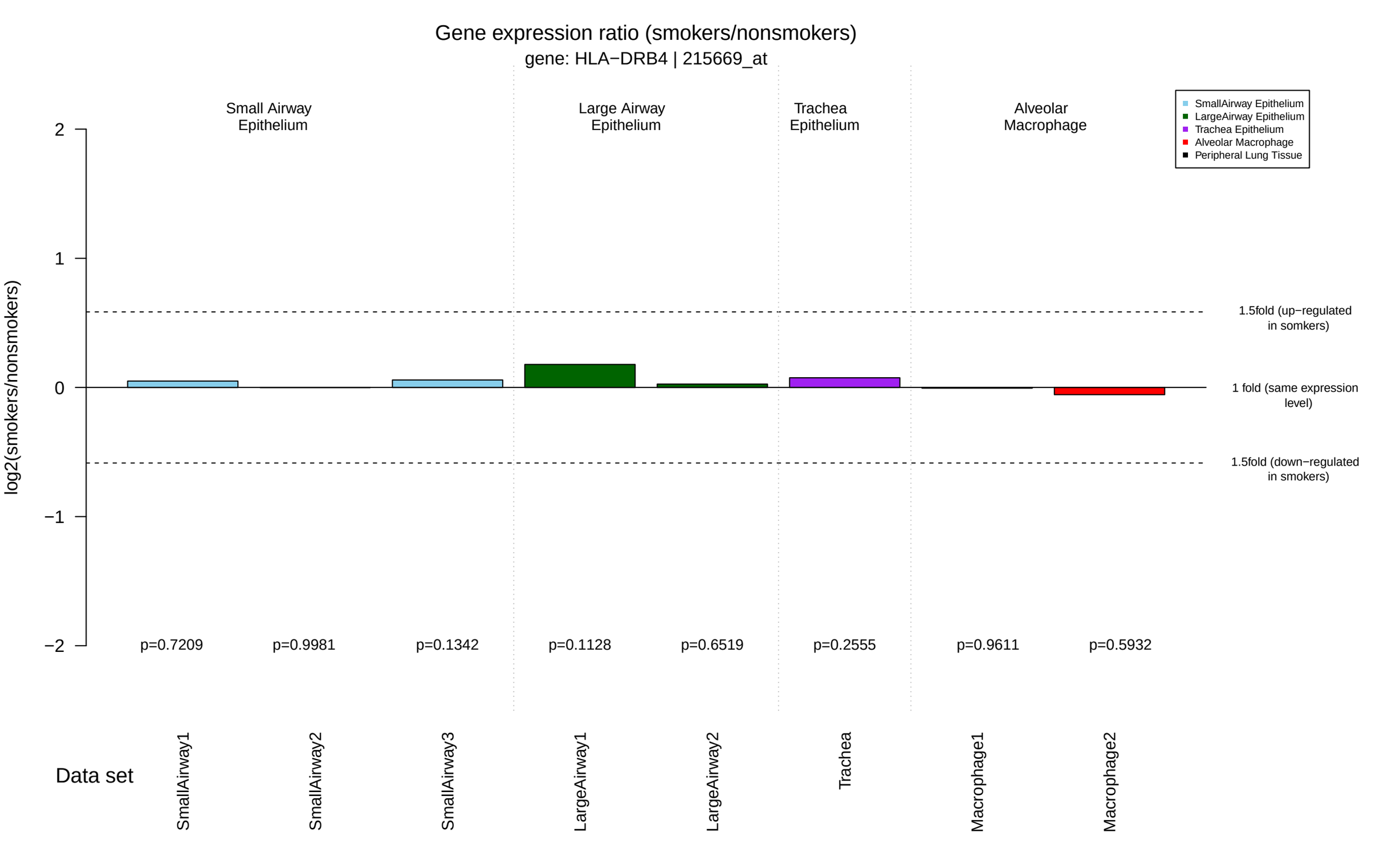
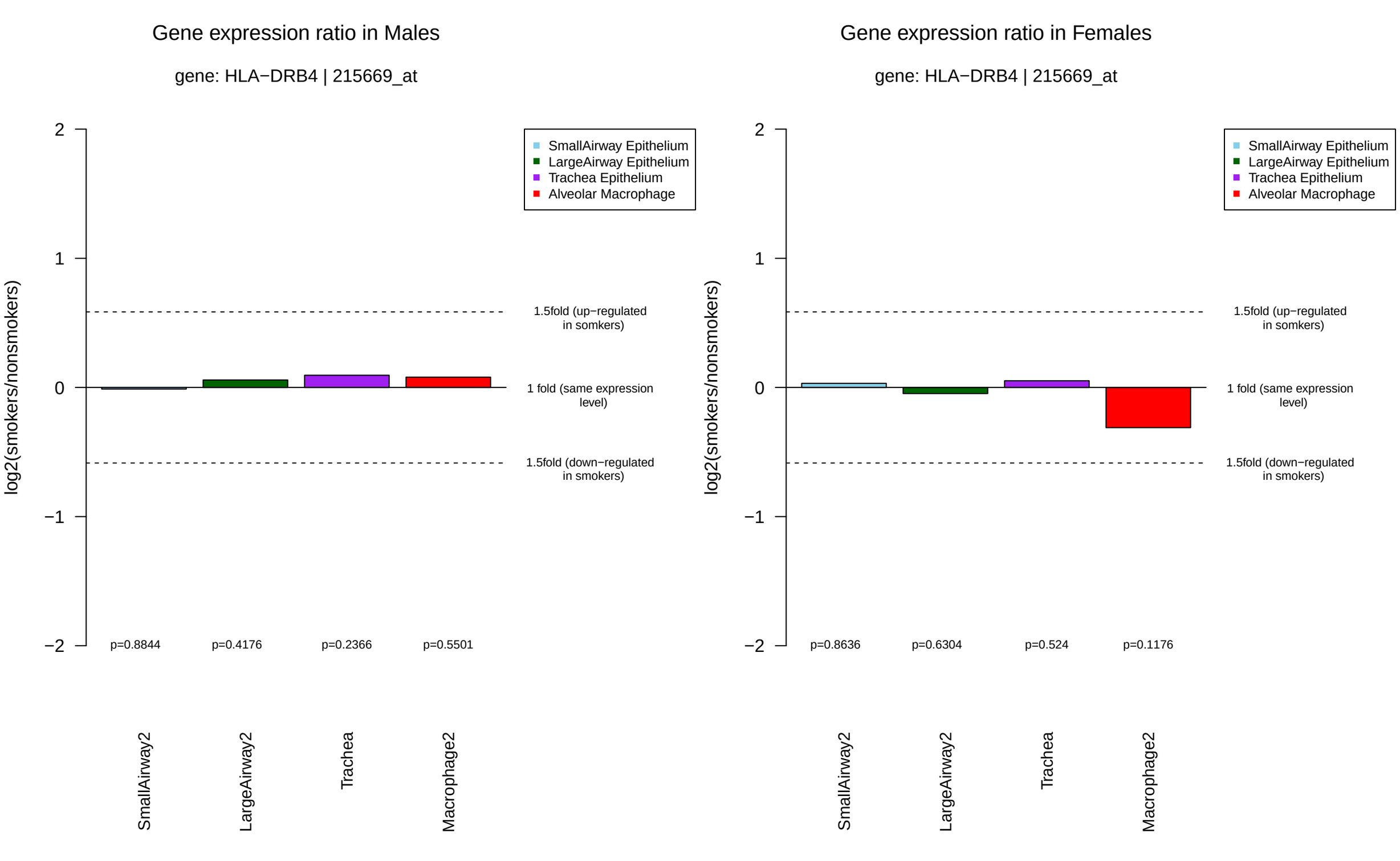
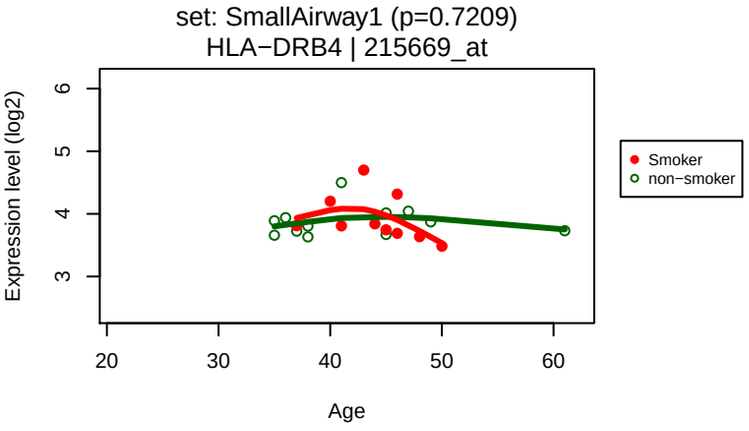
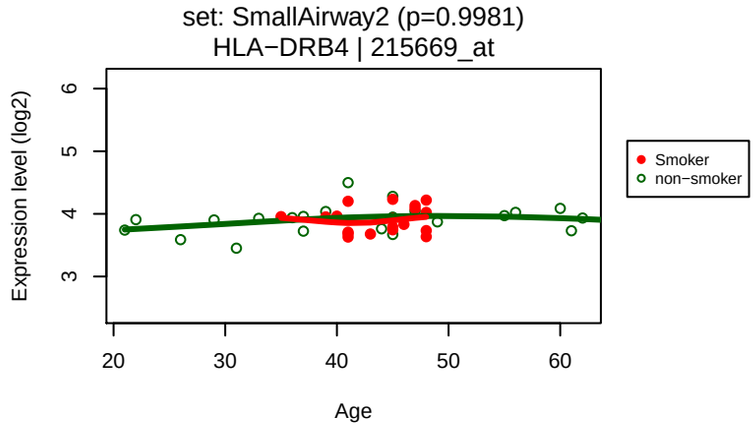
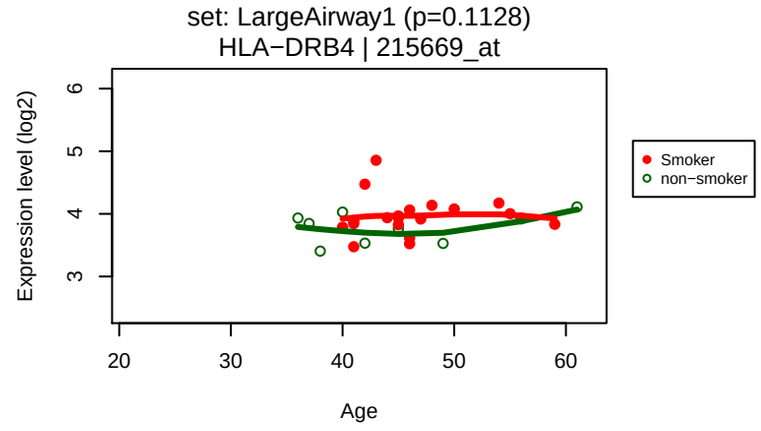

Gene expression of smokers and nonsmokers
gene: HLA−DRB4 | 215669_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7209
p=0.9981
p=0.1342
p=0.1128
p=0.6519
p=0.2555
p=0.9611
p=0.5932
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

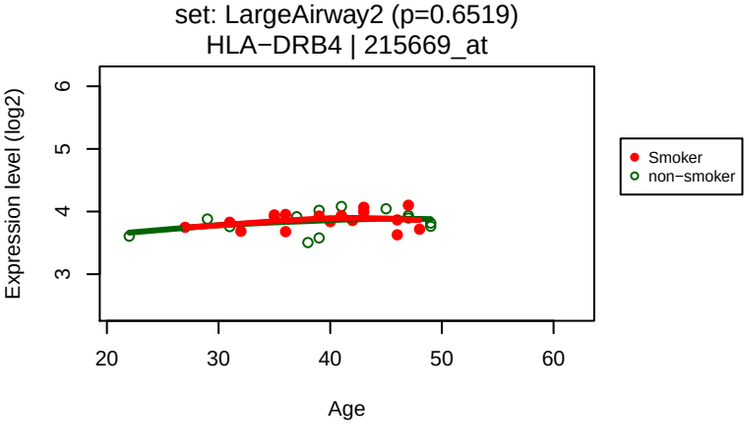

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRB4 | 215669_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7209
p=0.9981
p=0.1342
p=0.1128
p=0.6519
p=0.2555
p=0.9611
p=0.5932
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

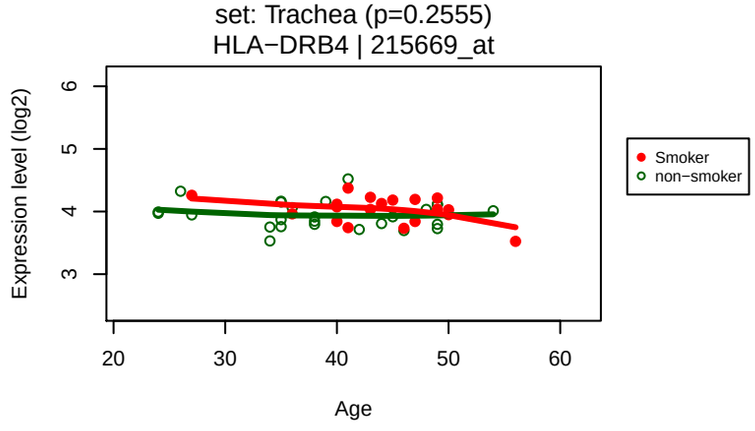

Gene expression ratio in Males
gene: HLA−DRB4 | 215669_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8844
p=0.4176
p=0.2366
p=0.5501
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DRB4 | 215669_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8636
p=0.6304
p=0.524
p=0.1176
SmallAirway2
LargeAirway2
Trachea
Macrophage2

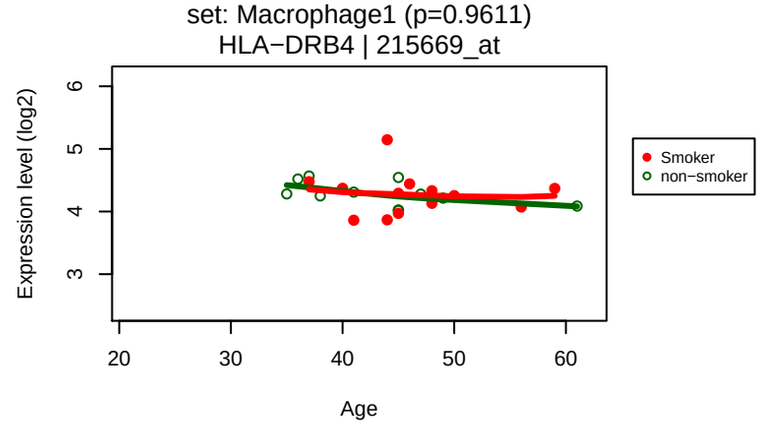

set: SmallAirway1 (p=0.7209)
HLA−DRB4 | 215669_at
Expression level (log2)
Age
Smoker
non−smoker

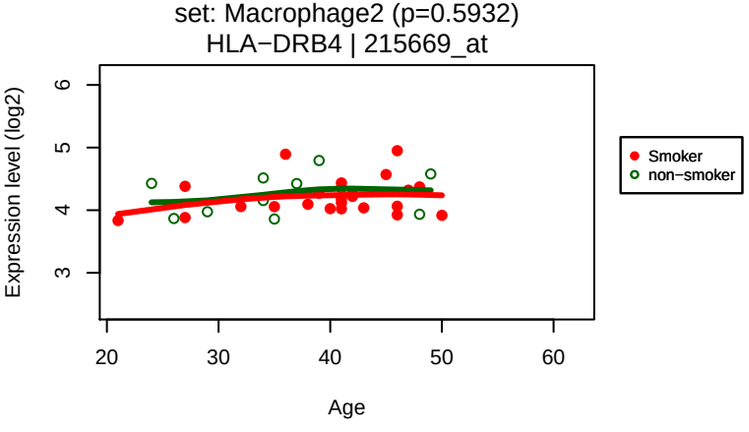

set: SmallAirway2 (p=0.9981)
HLA−DRB4 | 215669_at
Expression level (log2)
Age
Smoker
non−smoker

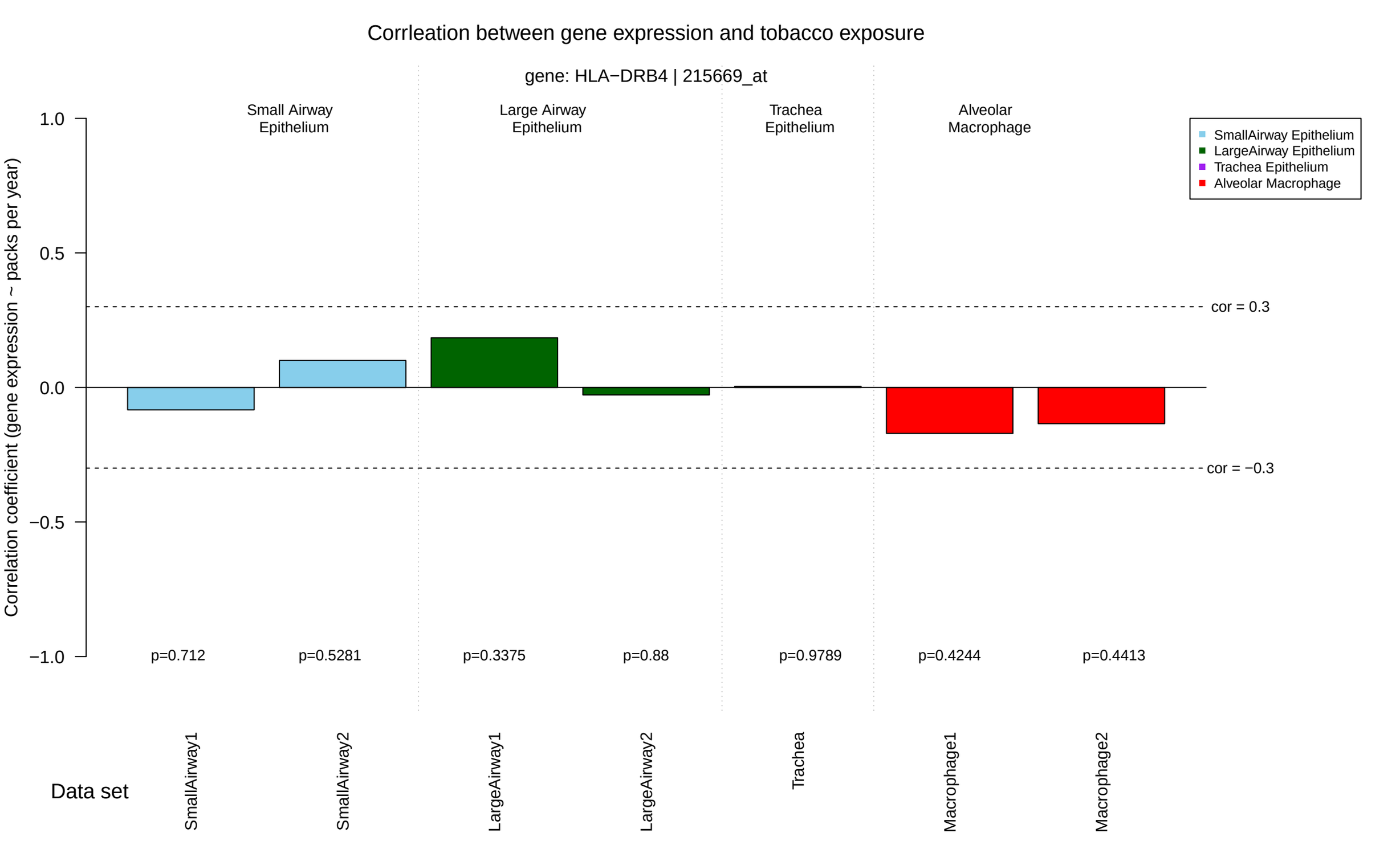

set: LargeAirway1 (p=0.1128)
HLA−DRB4 | 215669_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6519)
HLA−DRB4 | 215669_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2555)
HLA−DRB4 | 215669_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9611)
HLA−DRB4 | 215669_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5932)
HLA−DRB4 | 215669_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRB4 | 215669_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.712
p=0.5281
p=0.3375
p=0.88
p=0.9789
p=0.4244
p=0.4413
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2